Music Source Separation Using Structured Nonnegative Matrix Factorization

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Motivation

- Music source separation (MSS) recovers individual tracks from a single mixed recording observations
- **Problem:** Mixture collapses N sources into 1 channel → severely underdetermined
- **NMF** approximates the magnitude spectrogram **X ≈ WH**, where **W** is a spectral dictionary and **H** is temporal activations.
- Non-negativity encourages parts-based, interpretable decompositions aligned with how musical tones combine

Related Work

Shared Objective: $X \in \mathbb{R}_+^{F\times T}$ where $W \in \mathbb{R}_+^{F\times K}, H \in \mathbb{R}_+^{K\times T}$, such that $X \approx WH$

Method	Objective	H-update denominator
Euclidean NMF	$\|X - WH\|_F^2$	$W^\top WH + \varepsilon$
IS-NMF	$\sum_{f,t} d_{\mathrm{IS}}(X_{ft}\|[WH]_{ft})$	$W^\top \hat{X}^{-1} + \varepsilon$
Sparse NMF	$\|X - WH\|_F^2 + \lambda\|H\|_1$	$W^\top WH + \lambda + \varepsilon$
Conv. NMF	$\sum_{\tau=0}^{P-1} W_\tau H_{\leftarrow\tau} \approx X$	$\sum_\tau W_\tau^\top \mathbf{1} + \varepsilon$

IS divergence: $d_{IS}(x||y) = x/y - log(x/y) - 1$

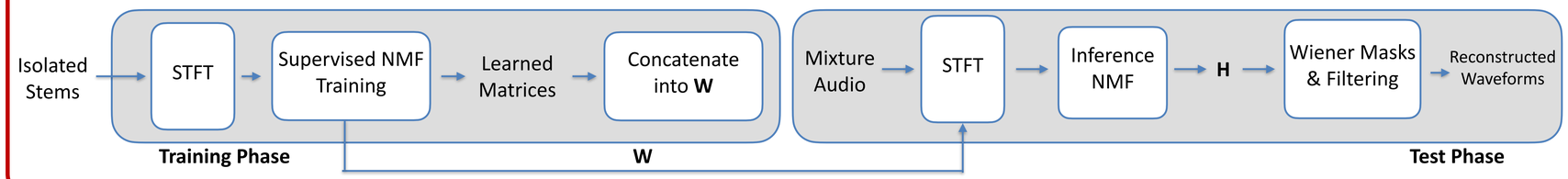
Wiener soft mask for source s: $M_{ft}^{(s)} = \dfrac{\sum_{k\in\mathcal{K}_s} W_{fk} H_{kt}}{\sum_{k=1}^{K} W_{fk} H_{kt} + \varepsilon}$

Multiplicative H update: $\mathbf{H} \leftarrow \mathbf{H} \odot \dfrac{\text{numerator}}{\text{denominator}}$

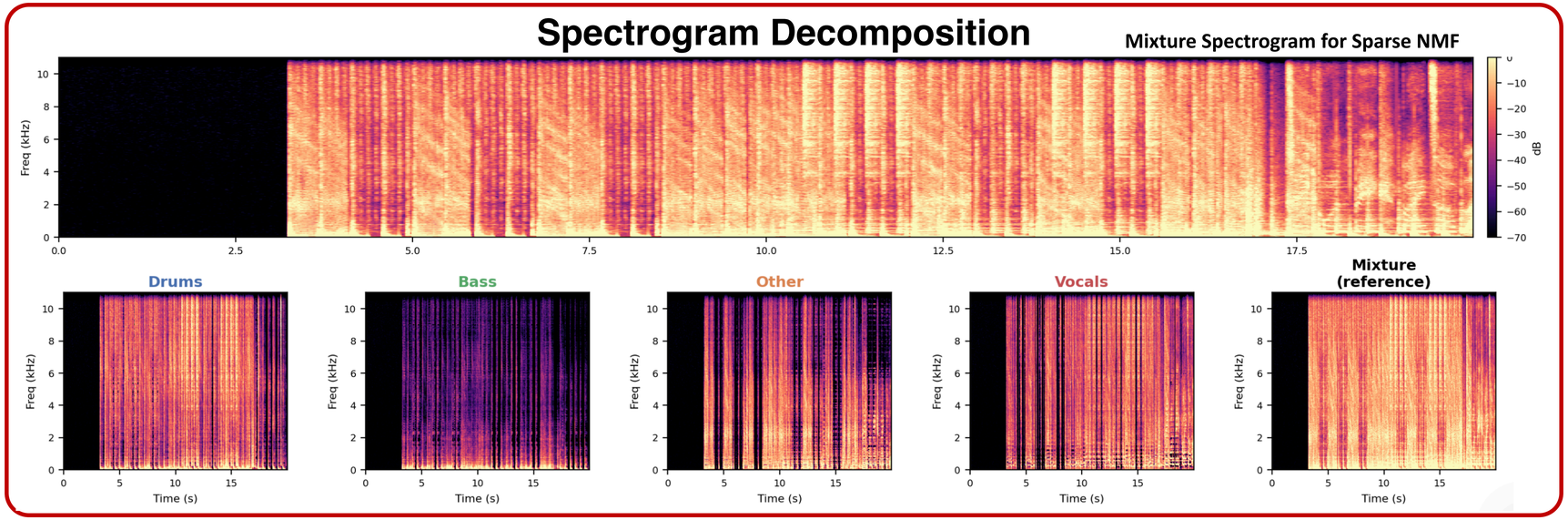
References

[1] Lee & Seung, "Algorithms for NMF," NeurIPS, 2000.
[2] Févotte et al., "NMF with the IS Divergence," Neural Comput., 2009.
[3] Smaragdis, "Non-negative Matrix Factor Deconvolution," ICA, 2004.
[4] Hoyer, "NMF with Sparseness Constraints," JMLR, 2004.
[5] Virtanen, "Monaural Sound Source Separation," IEEE TASLP, 2007.
[6] Rafii et al., "MUSDB18," Zenodo, 2017.
[7] Vincent et al., "BSS Eval Toolbox," IEEE TASLP, 2006.

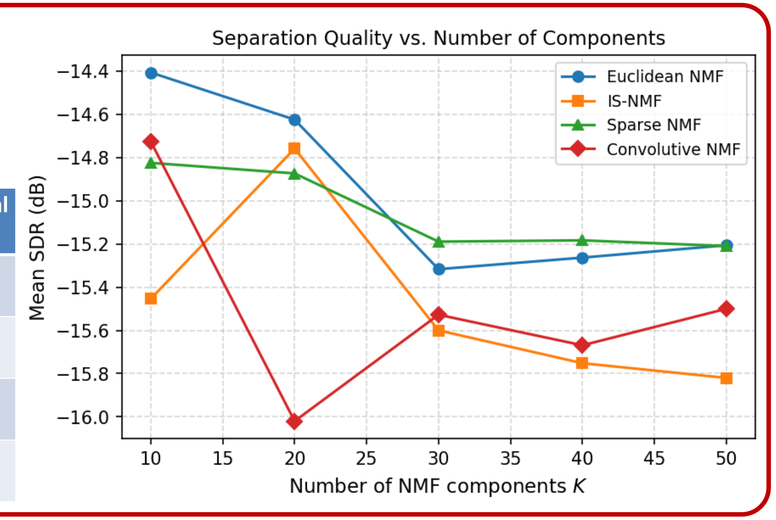
Supervised NMF Pipeline



Spectrogram Decomposition



BSS Evaluation Results

All methods are trained in a supervised setting on the MUSDB18 dataset and evaluated on 25 valid tracks from its 50-track test set using the BSS Eval metrics: SDR, SIR, and SAR.

Method	Vocals SDR	Bass SDR	Drums SDR	Other SDR	Computational Cost
Euclidean NMF	-22.1	-15.9	-3.1	0.4	26.0
IS-NMF	**-19.5**	-19.1	-3.1	**1.2**	0.6
Sparse NMF	-21.7	**-15.8**	**-2.9**	0.6	0.8
Conv. NMF	-19.6	-18.4	-6.8	0.6	0.3